Definite Integrals, Line Integrals, Work, Green-Type Thinking, and Vector Operations

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§1.1 Indefinite and definite integrals

Indefinite and definite integrals play different roles. An indefinite integral is an antiderivative family:

$$\int f(x)\,dx = F(x) + C, \qquad F'(x) = f(x).$$

A definite integral is a number:

$$\int_a^b f(x)\,dx.$$

Geometrically, it is signed area under the graph. Analytically, it is the limit of Riemann sums.

The fundamental theorem of calculus connects the two:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

when $F' = f$.

§1.2 Area by slicing

The page includes diagrams of shaded regions. The standard method is to slice a region into thin strips. If vertical strips are convenient, then

$$A = \int_a^b (y_{top}(x) - y_{bottom}(x))\,dx.$$

If horizontal strips are more convenient, then

$$A = \int_c^d (x_{right}(y) - x_{left}(y))\,dy.$$

Choosing the direction of slicing is part of the solution, not a fixed rule.

§1.3 Work as an integral

The physical meaning of a definite integral appears in work. If a force $F(x)$ acts along a line, then the work from $x = a$ to $x = b$ is

$$W = \int_a^b F(x)\,dx.$$

If the force is constant, this reduces to $W = F(b-a)$. If the force changes with position, integration adds the small contributions

$$dW = F(x)\,dx.$$

This is the same accumulation idea as area, but with a physical interpretation.

§1.4 Line integrals

The note then introduces line integrals. If a curve is parametrized by

$$r(t) = (x(t), y(t)), \qquad a \le t \le b,$$

then the scalar line element is

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2}\, dt.$$

A scalar line integral is

$$\int_C f\, ds = \int_a^b f(x(t), y(t))\sqrt{(x'(t))^2 + (y'(t))^2}\, dt.$$

For a vector field $F = (P, Q)$, the work integral is

$$\int_C F \cdot dr = \int_C P\, dx + Q\, dy = \int_a^b [P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)]\, dt.$$

§1.5 Green-type intuition

An oriented boundary and a vector field suggest that circulation around the boundary should be related to a double integral over the region. In the plane, Green's theorem says

$$\oint_{\partial D} P\, dx + Q\, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Even if the note does not prove the theorem fully, it records the key intuition: local rotation accumulated over an area produces circulation on the boundary.

§1.6 Polar coordinates

In polar coordinates, the change of variables is

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The area element becomes

$$dA = r\, dr\, d\theta.$$

The factor r is the Jacobian. Geometrically, a small polar rectangle has side lengths approximately dr and $r\, d\theta$.

Thus a disk of radius R has area

$$\int_0^{2\pi}\int_0^R r\, dr\, d\theta = \pi R^2.$$

§1.7 Spherical coordinates

For spherical coordinates, a common convention is

$$x = \rho\sin\varphi\cos\theta, \qquad y = \rho\sin\varphi\sin\theta, \qquad z = \rho\cos\varphi.$$

The volume element is

$$dV = \rho^2\sin\varphi\, d\rho\, d\varphi\, d\theta.$$

The factor $\rho^2\sin\varphi$ comes from the three small lengths:

$$d\rho, \qquad \rho\, d\varphi, \qquad \rho\sin\varphi\, d\theta.$$

This explains the diagram on the final page: the small spherical patch has dimensions determined by radius and angle.

§1.8 Dot product and cross product

The note also summarizes vector operations. The dot product is

$$a \cdot b = |a||b| \cos\theta.$$

It measures projection and appears in work:

$$dW = F \cdot dr.$$

The cross product has magnitude

$$|a \times b| = |a||b| \sin\theta$$

and direction given by orientation. It measures the area of the parallelogram spanned by a and b.

In coordinates,

$$a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

§1.9 After the examples: what object is being integrated?

The examples develop integration as accumulation over different geometric objects. The clean way to organize them is not by dimension alone, but by the type of object being integrated.

Integral	Object being integrated	Geometric meaning
$\int_a^b f(x)\,dx$	density on an oriented interval	signed accumulation
$\int_C f\,ds$	scalar density along a curve	mass/length-weighted accumulation
$\int_C P\,dx + Q\,dy$	one-form	work or circulation
$\iint_D f\,dA$	area density	accumulation over a region
$\iiint_\Omega f\,dV$	volume density	weighted volume or mass

This table prevents a common confusion. A line integral of $f\,ds$ and a work integral $P\,dx + Q\,dy$ both run along a curve, but they integrate different geometric objects. The first uses arclength and ignores orientation; the second is a pullback of a one-form and changes sign when the orientation is reversed.

Green's theorem then becomes a boundary-to-interior conversion:

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D (\partial_x Q - \partial_y P)\,dA.$$

It says that circulation around the boundary is curl accumulated over the interior.

§1.10 Line integrals as pullbacks

For a parametrized curve $\gamma : [a,b] \to \mathbb{R}^n$, a line integral of a one-form

$$\omega = P_1 dx_1 + \cdots + P_n dx_n$$

is

$$\int_\gamma \omega = \int_a^b \sum_i P_i(\gamma(t))\gamma_i'(t)\,dt.$$

This is the pullback of the one-form to the interval:

$$\gamma^*\omega = \sum_i P_i(\gamma(t))\gamma_i'(t)dt.$$

Thus the elementary formula $Pdx + Qdy$ is already differential-form notation.

§1.11 Exact forms and potentials

A one-form ω is exact if $\omega = df$ for some potential f. Then

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a)).$$

This is the fundamental theorem of line integrals.

A form is closed if $d\omega = 0$. In the plane, for $\omega = Pdx + Qdy$, this means

$$Q_x - P_y = 0.$$

On simply connected domains, closed one-forms are exact. On domains with holes, this may fail.

§1.12 Conservation laws

In mechanics, work is a line integral

$$W = \int_C F \cdot dr.$$

If $F = -\nabla V$ is conservative, then

$$W = V(A) - V(B)$$

depends only on endpoints. Conservation of energy is therefore a statement about exact one-forms.

This connects elementary work integrals to topology: path independence is not only about derivatives; it also depends on whether the domain has holes.

§1.13 De Rham preview

De Rham cohomology measures the failure of closed forms to be exact:

$$H^1_{dR}(M) = \frac{\{\text{closed 1-forms}\}}{\{\text{exact 1-forms}\}}.$$

For the punctured plane, the form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed but not exact. Its integral around the unit circle is 2π.

The elementary line integral around a circle is therefore a measurement of a topological hole.

§1.14 Work integrals and gauge freedom

If a force field is conservative, $F = -\nabla V$, then work depends only on endpoints. The potential V is not unique: adding a constant changes nothing. This is the simplest form of gauge freedom. The physically meaningful object is the differential dV, not the absolute zero of potential.

In electromagnetism, potentials have richer gauge freedom. A vector potential A may be changed by a gradient without changing the magnetic field $B = \nabla \times A$. The elementary fact that potentials are determined only up to constants is the first example of this broader principle.

§1.15 Polar and spherical adapted charts

The original integral examples using polar or spherical coordinates should be read as choosing charts adapted to symmetry. A chart is a coordinate map from a parameter domain to the geometric region. The integration element is obtained by pulling back the metric or volume form.

Thus polar coordinates are not merely a trick for circular regions. They are a coordinate chart on the punctured plane adapted to the action of rotations. Spherical coordinates are adapted to radial symmetry in three dimensions.